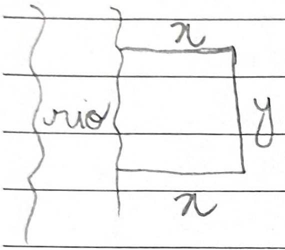
Otimização

Exemplo 1:



"perímetro": $p = 2x + y = 1200$

área: $A = x \cdot y$

$2x + y = 1200 \rightarrow y = 1200 - 2x$

$A = x(1200 - 2x) \rightarrow A = 1200x - 2x^2$

$\frac{dA}{dx} = 1200 - 4x = 0 \rightarrow 1200 - 4x = 0$

$//x = 300//$

$\frac{d^2A}{dx^2} = -4 \rightarrow$ ponto de máximo

$y = 1200 - 2 \cdot 300 = 600$

$$\boxed{\begin{array}{l} x = 300 \\ y = 600 \end{array}}$$

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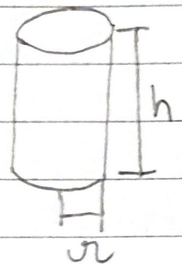
Exemplo 2



Área superficial total $\Rightarrow A = A_1 + 2A_2$

$$A_1 = 2\pi r h \quad , \quad A_2 = \pi r^2$$

$$A = 2\pi r h + 2\pi r^2$$

$$V = 1L = 1dm^3 = 1000\ cm^3$$

$$V = \pi r^2 . h \rightarrow h = \frac{1000}{\pi r^2}$$

$$A = \cancel{2\pi r} . \frac{1000}{\cancel{\pi r^2}} + 2\pi r^2 \rightarrow A = \frac{2000}{r} + 2\pi r^2$$

$$\frac{dA}{dr} = -2000 r^{-2} + 4\pi r \rightarrow \frac{dA}{dr} = -\frac{2000}{r^2} + 4\pi r$$

$$\frac{dA}{dr} = 0 \rightarrow -\frac{2000}{r^2} + 4\pi r = 0 \rightarrow -2000 + 4\pi r^3 = 0$$

$$4\pi r^3 = 2000 \rightarrow r^3 = \frac{500}{\pi} \rightarrow \boxed{r = \sqrt[3]{\frac{500}{\pi}}}$$

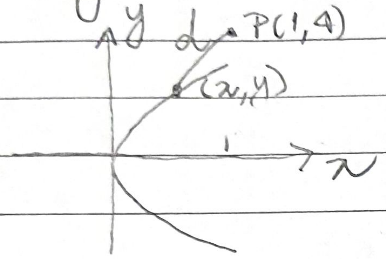
$$\frac{d^2A}{dr^2} = \frac{4000}{r^3} + 4\pi > 0 \rightarrow \text{ponto de mínimo}$$

$$h = \frac{1000}{\pi \left(\frac{500}{\pi}\right)^{2/3}} = \frac{1000^{3/3}}{\pi^{3/3}\ r^2} = \left(\frac{1000^3}{\pi^3}\right)^{1/3} \frac{1}{r^2} = \left[\frac{(2.500)^3}{\pi^3}\right]^{1/3} \frac{1}{r^2} = 2 . \frac{r^3}{r^2} = 2r$$

$$h = 2r \ , \text{ com } r = \sqrt[3]{\frac{500}{\pi}}$$

Exemplo 3:

$$y^2 = 2x$$



$$d = \sqrt{(x-1)^2 + (y-4)^2}$$

$$x = \frac{y^2}{2}$$

$$d = \sqrt{\left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2} \rightarrow d^2 = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2$$

$$(d^2)' = 2\left(\frac{y^2}{2} - 1\right) \cdot \frac{2y}{2} + 2(y-4) \cdot 1$$

$$(d^2)' = y^3 - 2y + 2y - 8 \rightarrow (d^2)' = y^3 - 8$$

$$(d^2)' = 0 \rightarrow y^3 - 8 = 0 \rightarrow y = 2 \,//$$

$$(d^2)'' = 3y^2 \rightarrow (d^2)''(2) = 12 > 0 \rightarrow$$ ponto de mínimo

$$y = 2 \;,\; x = 2 \qquad \boxed{P(2,2)} \,//$$

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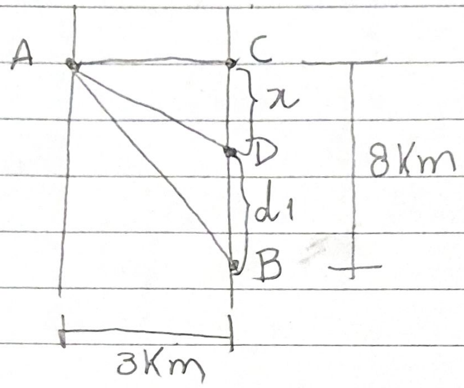
Exemplo 4



Distância caminhada: $d_1 = 8 - x$
Distância remada: d_2

Pelo Teorema de Pitágoras

$$d_2^2 = x^2 + 3^2$$

$$\text{Tempo} = \frac{\text{distância}}{\text{velocidade}} : \quad t_1 = \frac{d_1}{8} , \quad t_2 = \frac{d_2}{6}$$

$$t = t_1 + t_2 = \frac{d_1}{8} + \frac{d_2}{6} = \frac{8-x}{8} + \frac{\sqrt{x^2+9}}{6} \quad //$$

$$\frac{dt}{dx} = \frac{-1}{8} + \frac{1}{12\sqrt{x^2+9}} \cdot 2x = \frac{-1}{8} + \frac{x}{6\sqrt{x^2+9}}$$

$$\begin{array}{r} 64 \\ -36 \\ \hline 28 \end{array}$$

$$\frac{dt}{dx} = 0 \rightarrow \frac{x}{6\sqrt{x^2+9}} = \frac{1}{8} \rightarrow 8x = 6\sqrt{x^2+9} \rightarrow 64x^2 = 36x^2 + 36 \cdot 9$$

$$28x^2 = 36 \cdot 9 \xrightarrow{\div 4} 7x^2 = 9^2 \rightarrow x^2 = \frac{81}{7} \rightarrow x = \pm\frac{9}{\sqrt{7}}$$

$\boxed{x = 9/\sqrt{7}\ \text{Km}}$ // → mínimo absoluto

$$\text{Em } x = 0: \quad t = 1 + \frac{3}{6} = \frac{3}{2} ; \quad x = 8: \quad t = \frac{\sqrt{73}}{6} \approx 1,42$$

$$\begin{array}{r} 81 \\ +63 \\ \hline 154 \end{array} \qquad \begin{array}{l|l} 154 & 7 \\ 14 & 22 \\ 0 & \end{array}$$

$$\text{Em } x = \frac{9}{\sqrt{7}}\text{Km}: \quad t = \frac{8 - 9/\sqrt{7}}{8} + \frac{\sqrt{81/7 + 9}}{6} = 1 - \frac{9}{8\sqrt{7}} + \frac{\sqrt{154}}{6\sqrt{7}}$$

$$t = 1 - \frac{9}{8\sqrt{7}} + \frac{\sqrt{22}}{6} \approx 1 - 0,425 + 0,782 \approx 1,3 \quad //$$

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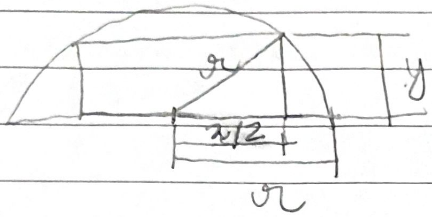
Exemplo 5



$A = x \cdot y$

$r^2 = y^2 + \frac{x^2}{4} \Rightarrow y = \sqrt{r^2 - \frac{x^2}{4}}$

$$A = x \cdot \sqrt{r^2 - \frac{x^2}{4}}$$

$$\frac{dA}{dx} = \sqrt{r^2 - \frac{x^2}{4}} + x \cdot \frac{1}{2} \frac{(-x/2)}{\sqrt{r^2 - \frac{x^2}{4}}}$$

$$\frac{dA}{dx} = \sqrt{r^2 - x^2/4} - \frac{x^2}{4\sqrt{r^2 - x^2/4}} = \frac{4r^2 - x^2 - x^2}{4\sqrt{r^2 - x^2/4}}$$

$$\frac{dA}{dx} = \frac{2r^2 - x^2}{2\sqrt{r^2 - x^2/4}} = 0 \rightarrow x^2 = 2r^2 \Rightarrow // x = \sqrt{2}\, r //$$

$x = 0,\ A = 0 //$ $x = r,\ y = 0$ e $A = 0$

$$y = \sqrt{r^2 - \frac{2r^2}{4}} = \sqrt{r^2 - \frac{r^2}{2}} = \sqrt{\frac{r^2}{2}} = \frac{r}{\sqrt{2}} = \frac{r\sqrt{2}}{2} //$$

$$A = r\sqrt{2} \cdot \frac{r\sqrt{2}}{2} = r^2 //$$

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Exemplo 6:

x = número de aparelhos vendidos por semana

↳ Aumento semanal: $x - 200$//

Para cada 20 unidades vendidas o preço cai em R$10,00

↳ Para cada unidade adicional, o decréscimo no preço será $1/20 \cdot 10$

Função demanda:

$$p(x) = 350 - (x-200) \cdot \frac{10}{20} = 350 - \frac{x}{2} + 100$$

$$p(x) = 450 - \frac{x}{2}$$

Função receita

$$R(x) = x \cdot p(x) = 450x - \frac{x^2}{2}$$

$R'(x) = 450 - x \rightarrow R'(x) = 0 \rightarrow$ // $x = 450$//

$p(450) = 225$//

Desconto $350 - 225$ = R$ 125,00//

$$\begin{array}{r} 350 \\ -225 \\ \hline 125 \end{array}$$

spiral®

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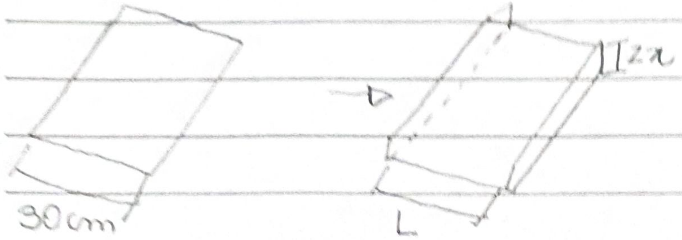
Exemplo 7:



$L = 30 - 2x$

Capacidade máxima: $x(30-2x)$

$A = x(30-2x) = 30x - 2x^2$

$A' = 30 - 4x = 0 \rightarrow x = \frac{30}{4} = \frac{15}{2}$ //

$x_{min} = 0 \rightarrow A = 0$

$x_{máx} = 15 \rightarrow A = 0$

$x^* = \frac{15}{2} \rightarrow A = \frac{15}{2}(30-15) = \frac{225}{2}$ //

$$\begin{array}{r} 15 \\ \times 15 \\ \hline 75 \\ 150 \\ \hline 225 \end{array}$$

$A'' = -4 \rightarrow$ máximo

Deve-se dobrar $x = 15/2$ de cada lado

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$$\begin{array}{r} 52 \\ \times 40 \\ \hline 2080 \end{array}$$

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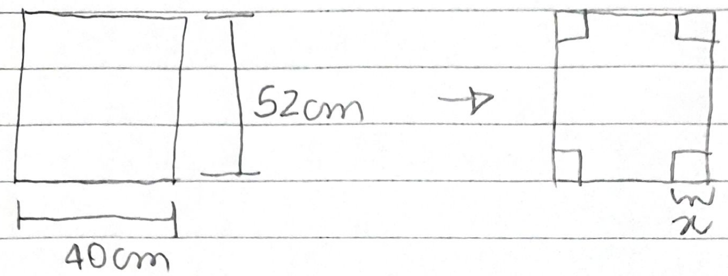
$$\begin{array}{r} 184 \\ \times\ 2 \\ \hline 378 \end{array}$$

Exemplo 8:



Base $b = 40 - 2x$
Comprimento: $c = 52 - 2x$
Altura: $h = x$

Volume: $V = x(40-2x)(52-2x)$, $0 \leq x \leq 20$

$V = x(2080 - 80x - 104x + 4x^2) = 4x^3 - 184x^2 + 2080x$ //

$V' = 12x^2 - 378x + 2080 = 0 \rightarrow 6x^2 - 184x + 1040 = 0$

$$\begin{array}{r|l} 2224 & 2 \\ 1112 & 2 \\ 556 & 2 \\ 278 & 2 \\ 139 & \end{array}$$

$3x^2 - 92x + 520 = 0$

$$x = \frac{92 \pm \sqrt{8464 - 6240}}{6} = \frac{92 \pm \sqrt{2224}}{6} = \frac{92 \pm 4\sqrt{139}}{6}$$

$$x = \frac{46 \pm 2\sqrt{139}}{3} \qquad x_1 \approx 23,19 \ , \ x_2 \approx 7,47$$

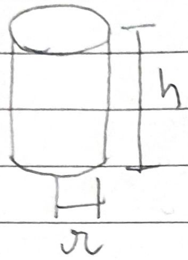
$V'' = 6x - 92$

$V''(23,19) = 47,14$; $V''(7,47) = -47,18$

↳ máximo

// $x = 7,47$ cm //

Exemplo 9



$$V = 375\pi \, cm^2 = \pi r^2 h$$

$$\left.\begin{array}{l} A_b = \pi r^2 \\ A_s = 2\pi r . h \end{array}\right\} A_T = \pi r^2 + 2\pi r h$$

Custo: $p = 15 . \pi r^2 + 5 . 2\pi r h = 15\pi r^2 + 10\pi r h$

$$\pi r^2 h = 375\pi \rightarrow h = \frac{375}{r^2}$$

$$p = 15\pi r^2 + 10\pi r . \frac{375}{r^2} \rightarrow p = 15\pi r^2 + \frac{3750\pi}{r}$$

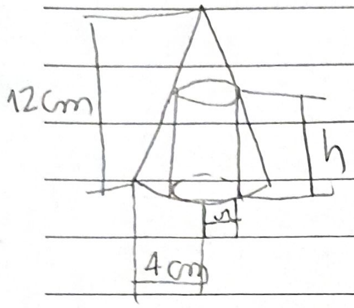
$$p' = 30\pi r - \frac{3750\pi}{r^2} = 0 \rightarrow 30\pi r^3 - 3750\pi = 0$$

$$r^3 = \frac{3750}{30} = 125 \rightarrow \boxed{r = 5}$$

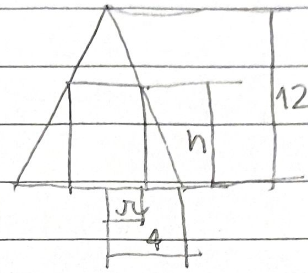
$p'' = 30\pi + \frac{7500\pi}{r^3} > 0 \rightarrow$ ponto de mínimo //

// $r = 5$ cm , $h = \frac{375}{25} = 15$ cm //

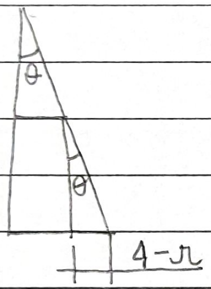
Exemplo 10:



$\rightarrow$



Volume do cilindro
$V = \pi r^2 . h$ //



$$tg\,\theta = \frac{4-r}{h} \Rightarrow \frac{r}{12-h} = \frac{4}{12}$$

$$\frac{4-r}{h} = \frac{4}{12} \rightarrow 4-r = \frac{h}{3} \rightarrow \boxed{h = 3(4-r)}$$

$$V = \pi r^2 . 3(4-r) \rightarrow V = 3\pi(4r^2 - r^3) = 3\pi r^2 (4-r)$$

$r\ min = 0 \rightarrow V = 0$

$r\ máx = 4 \rightarrow V = 0$

$$V' = 3\pi(8r - 3r^2) = 0 \rightarrow 3r^2 - 8r = 0 \rightarrow r = 0$$
$$\rightarrow r = 8/3$$

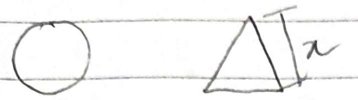
$$V'' = 3\pi(8 - 6r) \rightarrow V''(8/3) = 3\pi(8-16) = -24\pi < 0 \rightarrow \text{ponto de máximo}$$

$$h(8/3) = 3\left(4 - \frac{8}{3}\right) = 3 . \frac{4}{3} = 4 \text{ //}$$

$$V = \pi . \frac{64}{9} . 4 = \frac{256\pi}{9} \text{ cm}^3$$

Exemplo 11:

$$p_1 + p_2 = 1{,}5$$



$p_1 = 2\pi r$ $\quad$ $p_2 = 3x$

$$3x = 1{,}5 - 2\pi r$$

$$A_1 = \pi r^2 , \quad A_2 = \frac{x^2\sqrt{3}}{4}$$

$$3x = 1{,}5 - 2\pi r \rightarrow r = \frac{1{,}5 - 3x}{2\pi}$$

$$A_T = \pi r^2 + \frac{x^2\sqrt{3}}{4} = \pi\left(\frac{1{,}5-3x}{2\pi}\right)^2 + \frac{x^2\sqrt{3}}{4}$$

$$A_T' = \cancel{\pi}\cdot\cancel{2}\left(\frac{1{,}5-3x}{\cancel{2\pi}}\right)\frac{(-3)}{2\pi} + \frac{x\sqrt{3}}{2} = \frac{9x - 4{,}5}{2\pi} + \frac{x\sqrt{3}}{2} = 0 \quad \times 2$$

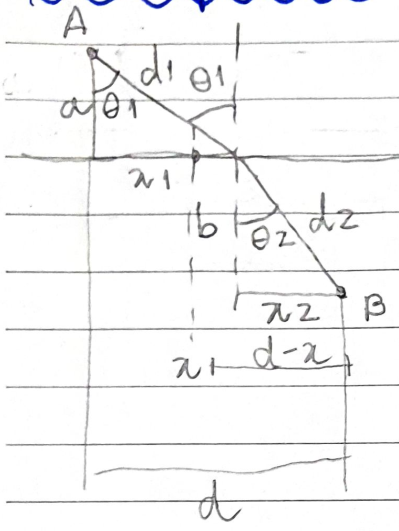
$$A_T' = \frac{9x}{\pi} + x\sqrt{3} - \frac{4{,}5}{\pi} = 0 \rightarrow x\left(9/\pi + \sqrt{3}\right) = \frac{4{,}5}{\pi} \rightarrow x \simeq \frac{1{,}433}{4{,}598}$$

$x = 0{,}3116$ $\quad$ $p_2 = 0{,}935$

$p_1 = 0{,}565$

$$A_T'' = \frac{9}{\pi} + \sqrt{3} \rightarrow \text{mínimo}$$

Exemplo 12:



$$t_1 = \frac{d_1}{c_1}, \quad t_2 = \frac{d_2}{c_2}$$

$$d_1 = \sqrt{x^2+a^2}$$

$$d_2 = \sqrt{(d-x)^2+b^2}$$

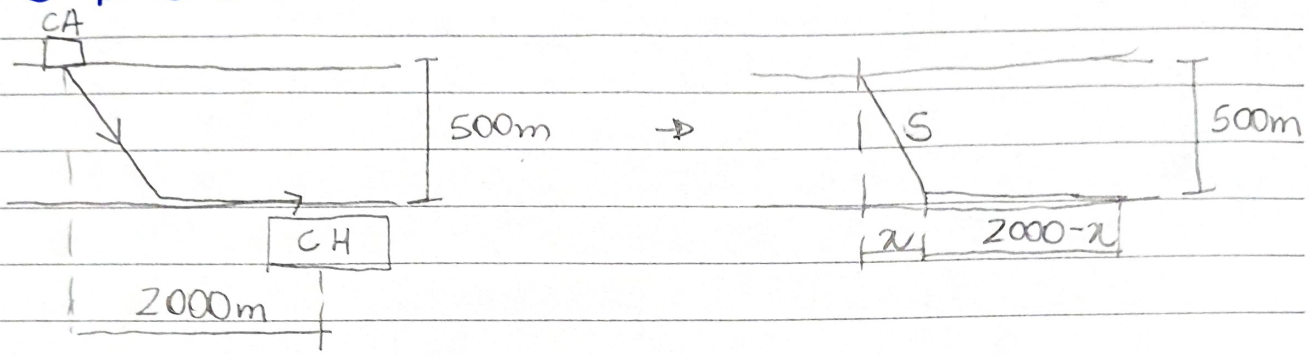
$$t = t_1 + t_2 = \frac{\sqrt{x^2+a^2}}{c_1} + \frac{\sqrt{(d-x)^2+b^2}}{c^2}$$

$$\frac{dt}{dx} = \frac{1}{\cancel{2}c_1}\frac{\cancel{2}x}{\sqrt{x^2+a^2}} + \frac{1}{\cancel{2}c_2}\frac{\cancel{2}(d-x)(-1)}{\sqrt{(d-x)^2+b^2}}$$

$$\frac{dt}{dx} = \frac{x}{c_1\sqrt{x^2+a^2}} - \frac{(d-x)}{c_2\sqrt{(d-x)^2+b^2}} = \frac{x_1}{c_1 . d_1} - \frac{x_2}{c_2 . d_2} = \frac{\operatorname{sen}\theta_1}{c_1} - \frac{\operatorname{sen}\theta_2}{c_2}$$

$$\frac{dt}{dx} = 0 \text{ se } \frac{\operatorname{sen}\theta_1}{c_1} = \frac{\operatorname{sen}\theta_2}{c_2} \;//$$

Exemplo 13



$$S^2 = 500^2 + x^2 \Rightarrow S = \sqrt{500^2+x^2}$$

$$p = 640 \cdot s + 312 \cdot (2000-x) = 640\sqrt{500^2+x^2} + 312(2000-x)$$

$$p' = \frac{640}{\cancel{2}} \cdot \frac{\cancel{2}x}{\sqrt{500^2+x^2}} - 312 = 0 \Rightarrow x = \frac{312}{640} \cdot \sqrt{500^2+x^2}$$

$$x^2 = \underbrace{\left(\frac{312}{640}\right)^2}_{0,2376} \cdot x^2 + \left(\frac{312}{640}\right)^2 \cdot 500^2 \Rightarrow x \approx \sqrt{\frac{0,237 \cdot 500^2}{0,763}}$$

$$x \approx 279$$

$$p'' = \frac{640\sqrt{500^2+x^2} - 640 \cdot \frac{1}{\cancel{2}}(500^2+x^2)^{-1/2} \cdot \cancel{2}x}{500^2+x^2}$$

$$p'' = \frac{640(500^2+x^2)^{1/2} - 640x(500^2+x^2)^{-1/2}}{500^2+x^2} = \frac{640(500^2+x^2) - 640x}{(500^2+x^2)} > 0$$

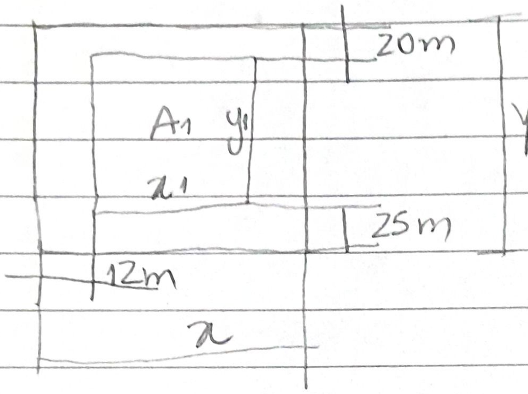
em $x \approx 279$, temos um mínimo: $p = 903.399,36$

$x = 0 \Rightarrow p = 944.000$

$x = 2000 \Rightarrow p = 1.319,393$

$x = 279$ é mínimo absoluto

Exemplo 14:



$A_1 = 12100\, m^2 = x_1 y_1$

Área do lote: $A = x.y$

$x = x_1 + 24$

$y = y_1 + 45$

$A = (x_1 + 24)(y_1 + 45)$ $\qquad y_1 = \frac{12100}{x_1}$

$A = (x_1 + 24)\left(\frac{12100}{x_1} + 45\right)$

$A = 12100 + 45x_1 + \frac{290400}{x_1} + 69 \Rightarrow A = 45x_1 + \frac{290400}{x_1} + 12169$

$A' = 45 - \frac{290400}{x_1^2} = 0 \rightarrow x_1^2 = \frac{290400}{45} \Rightarrow x_1 \approx 80,33m$

Então $y_1 = \frac{12100}{x_1} \approx 150,63m$

Assim, o lote tem as dimensões

$x = 104,33m$

$y = 195,63m$

e área $A = 20\,410,08m^2$